# IPL Performance Analytics Report

A Visualization of Team Efficiency, Player Impact and Match Dynamics

Student Name: Chalodiya Utsav Hareshbhai

Student ID: 25042720

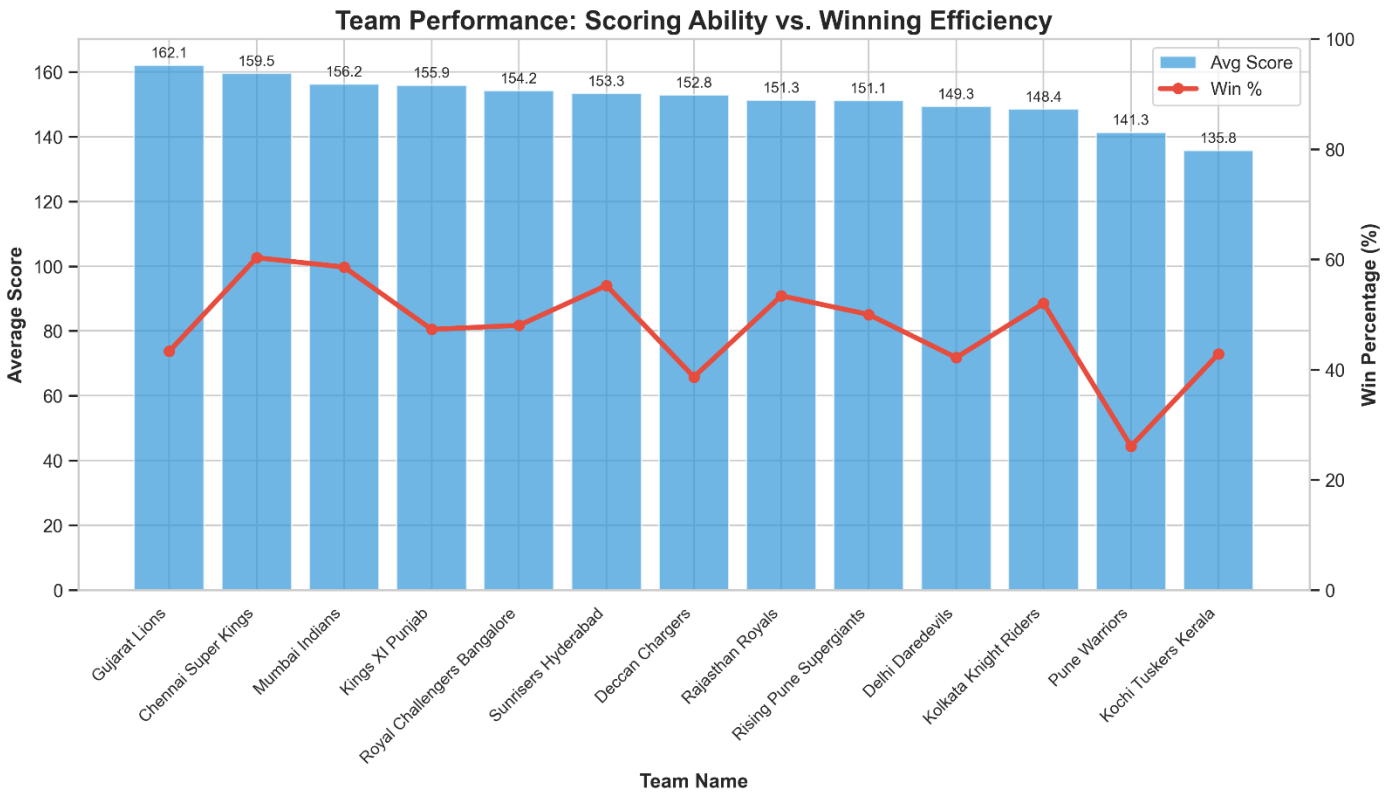
## 1. Introduction

The Indian Premier League (IPL) is one of the most competitive T20 cricket league. In cricket, match result is influenced by multiple factors ana player performance such as batting efficiency, bowling power, and strategic driven decision. Main statics like total runs or wickets often fail to explain the result or complete performance of the team.

This report shows data-driven analytical techniques to interpret IPL performance across all factors. This analytical study includes match level and ball-by-ball datasets to identify trends and patterns in team success, player contributions, toss decision and more. The goal of the reports is to develop performance indicators that explain match outcomes better.

## 2. Visual Analysis

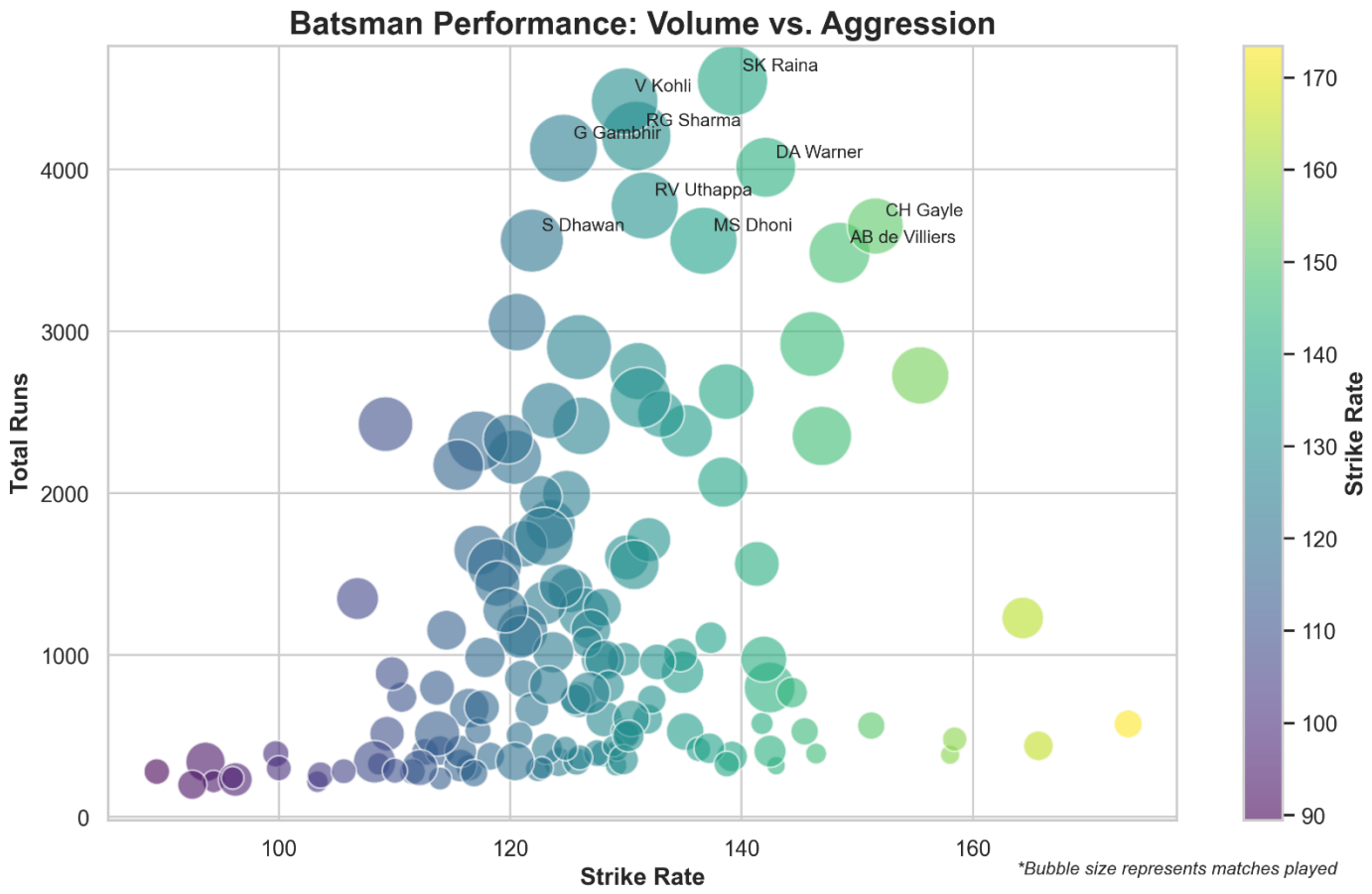
**Team Performance:**




The comparison between average scores and win percentages visual depicts that higher scoring does not mean that higher winning percentage. Such as Gujarat Lions had high scoring average but moderate win rates. Whereas team like Chennai Super Kings and Mumbai Indian had a strong win consistency despite of lower average scores than Gujarat Lions.

This describes that match efficiency and game management are more essential than scoring ability.

**Batsman Performance Analysis:**



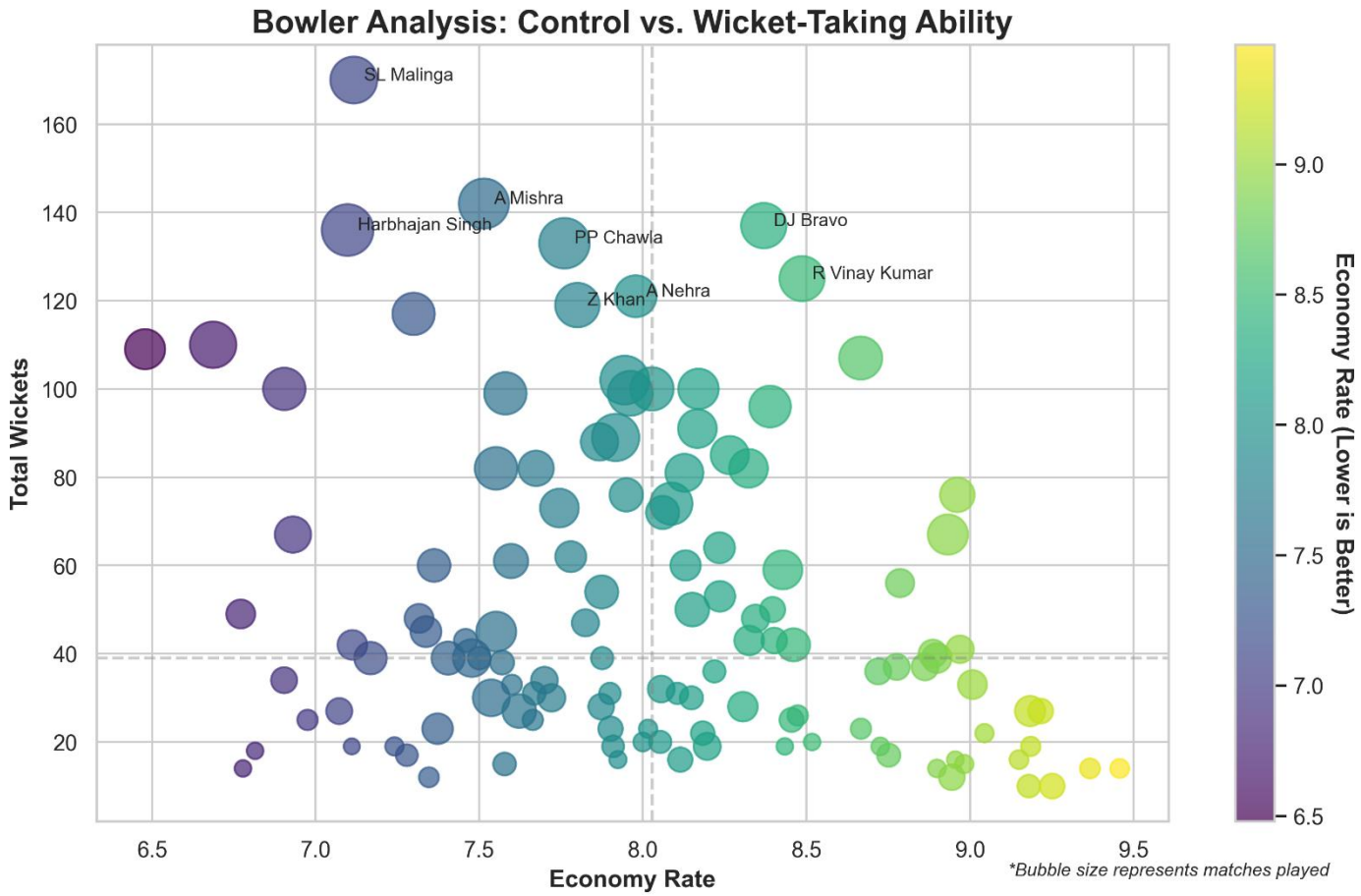

Batsman performance is evaluated using total runs(volume) and strike-rate(aggression). Visual suggests the insight of batsman ability to score that high-volume player provide consistent in scoring where high-strike rate players give high-scoring result but not consistency. A very few players successfully balance that both dimensions.

The most effective batsman is those who stabilize in both runs and scoring speed, got success and help their team towards the win.

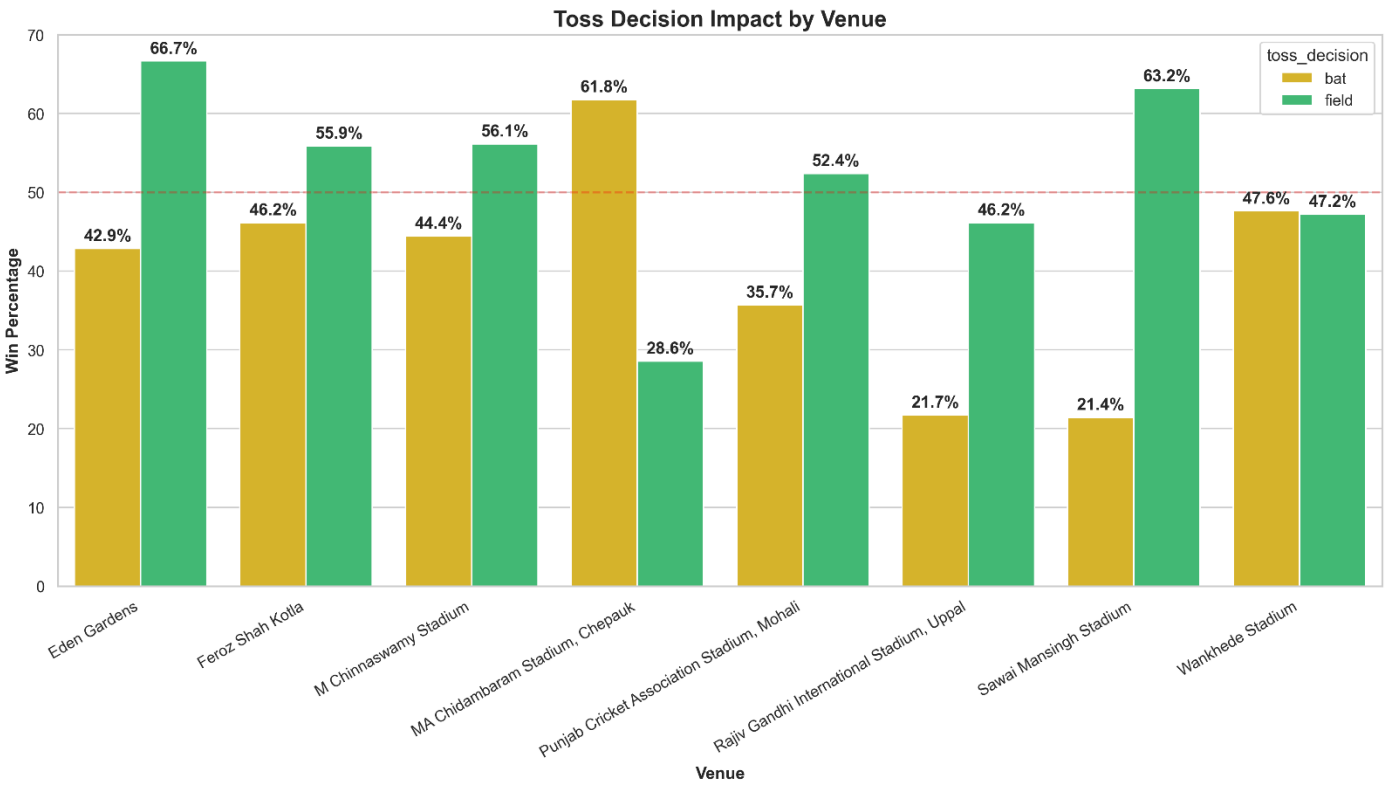
**Bowler Performance Analysis:**

Bowler effectiveness is analysed by the two important factors of bowler, economy rate and total wickets. In bowler side the visual said that it didn't follow the batting side trend, here Elite bowlers achieve both it factor, low economy rate and high wicket counts. Perhaps there are some bowlers who gave excessive runs but take wickets also. Where some of bowlers control their economy but they can't succeed to take wickets.

In bowling success depends in balancing runs economy and wicket taking ability, rather than going in only one direction.




## Impact of Toss Decisions:



This impact of toss decisions was across all major venues. The visual suggest that fielding first give an advantage at venue like Eden Garden. Batting first is more successful at venues Chepauk. Hence in major stadium fielding first give an advantage for toss decision team to win match. But there is no universal strategy applies in across all venues. It all depends on the venue and the pitch of the venue.

Match outcomes are not influenced by only toss decision it has some other factors which influenced in over all matches such as pitch conditions and weather factor.

## 3. Table Analysis

### Team Efficiency and Pressure Performance

**Team Efficiency & Pressure Performance Report**

| rank | team | avg_score | pressure_avg_runs | win_percentage | efficiency_index |
|---|---|---|---|---|---|
| 1 | Mumbai Indians | 156.18 | 47.0 | 58.6 | 0.38 |
| 1 | Chennai Super Kings | 159.53 | 46.66 | 60.31 | 0.38 |
| 2 | Sunrisers Hyderabad | 153.32 | 44.97 | 55.26 | 0.36 |
| 3 | Kolkata Knight Riders | 148.41 | 41.53 | 52.03 | 0.35 |
| 3 | Rajasthan Royals | 151.31 | 40.72 | 53.39 | 0.35 |
| 4 | Rising Pune Supergiants | 151.1 | 46.58 | 50.0 | 0.33 |
| 5 | Kochi Tuskers Kerala | 135.79 | 35.45 | 42.86 | 0.32 |
| 6 | Royal Challengers Bangalore | 154.18 | 47.01 | 48.03 | 0.31 |
| 7 | Kings XI Punjab | 155.86 | 43.4 | 47.3 | 0.3 |
| 8 | Delhi Daredevils | 149.34 | 40.92 | 42.18 | 0.28 |
| 9 | Gujarat Lions | 162.07 | 41.39 | 43.33 | 0.27 |
| 10 | Deccan Chargers | 152.84 | 43.51 | 38.67 | 0.25 |
| 11 | Pune Warriors | 141.29 | 39.39 | 26.09 | 0.18 |

A table was developed with so many factors and columns such as, average score, pressure (death overs) performance, win percentage and efficiency index. Mumbai Indians and Chennai Super Kings at a top on the table in the efficiency and also the have a high average runs in pressure. Were lower ranked team shows inefficiency in pressure situations.

Performance under pressure is a critical factor between successful and unsuccessful team.

## 4. Conclusion

The Analysis suggests that success in the IPL cricket is not only determined by scoring ability or an individual performance. Instead of this, it is affected by balanced contributions of batsmen and bowlers, strategic decision during the match by understanding the factor that affects the match and strong performance under pressure.